

Yasha Duncan-Wilson

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Education

Heriot-Watt University – MEng in Aerospace Engineering

2028

Experience

Mechanical Engineer Heriot-Watt University CanSat Society – Edinburgh 2023 – Present

- Complied with challenging mass budget by reducing mass of parts in the flight vehicle through novel manufacturing techniques, leading to a 75% reduction in mass while meeting strength requirements.
- Adapted aerospace industry standard design patterns for hobbyist scale while meeting stringent power and cost requirements, such that the vehicle survived testing expected to be destructive.
- Performed analyses of competing solutions to complex problems, ultimately choosing designs which were fit for purpose without excessive mass or fragility.

Freelance Software Engineer Waverley Management Consultants Interactive Futures Scanner – Edinburgh 2021

- Produced interactive Horizon Scanning tool using React.JS and Python
- Designed front-end to match existing design scheme as closely as possible while elegantly integrating interactivity
- Produced back-end software for adapting existing scans to front-end.
- Client shelved project before completion.

Projects

Ender 3 Klipper Conversion <https://github.com/mechanical-girl/klipper-ender-3>

- Improved poor print success rates: built and fitted an ABL probe to improve first-layer reliability
- Upgraded firmware: wrote and tuned Klipper config to improve print outcomes and user experience

Skills

Programming: Python, React.JS

3D Printer Firmware: Marlin, Klipper

CAD: Solidworks, Fusion 360